MIL-B-52817A(ME)
20 June 1979
SUPERSEDING
MIL-B-52817(ME)
25 June 1974

MILITARY SPECIFICATION

BRACKET, SCAFFOLD, FOLDING

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a folding scaffold bracket with an attached handrail adapter bracket.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-B-575	- Bolts, Hexagon and Square.
FF-N-836	- Nut: Square, Hexagon, Cap, Slotted, Castle, Knurled, Welding and Single Ball Seat.

FSC 5440

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:34Z
Check the source to verify that this is the current version before use.

WW-P-406	- Pipe, Steel (Seamless and Welded) (For Ordinary Use).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MS27040	- Nut, Plain, Square - Steel, Cadmium Plated.
MS35355	- Bolt, Machine, Square Head, Steel, Cadmium or Zinc Plated, UNC-2A.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 506 - Regular Quality Hot-Rolled and Cold-Rolled Alloy-Steel Sheets and Strip.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:34Z
Check the source to verify that this is the current version before use.

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The bracket shall be as shown in Figures 1 and 2 and as specified herein.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.3).

3.2.1 Sheet steel. Sheet steel shall conform to ASTM A 506, sheet, hot rolled, commercial quality, cut edge.

3.2.2 Pipe. Steel pipe shall conform to WW-P-406, Weight A, Class 1, unthreaded with square cut ends.

3.2.3 Bolts. Square head bolts used in the assembly of the folding scaffold bracket shall conform to MS35355-124, material in accordance with FF-B-575. The square head bolts used to assemble the handrail adapter bracket to the folding scaffold bracket shall conform to MS35355-126, material in accordance with FF-B-575.

3.2.4 Nuts. Nuts shall conform to MS27040-14, material in accordance with FF-N-836.

3.3 Treatment and painting. The portions of the bracket normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A, color as specified (see 6.2).

3.4 Bracket assembly. The bracket will be assembled as shown in Figure 1. The handrail adapter bracket shall not be assembled to the scaffold bracket; however, the 1.25-inch square head bolt with nut will be placed into the 0.5625 inch drilled hole in this bracket prior to shipment.

3.5 Workmanship.

3.5.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:34Z
Check the source to verify that this is the current version before use.

All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.5.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the bracket to conform to the examination and test requirements specified in Section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 0.187 inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2) the welding process used in fabrication of the bracket shall be at the option of the contractor.

3.5.3 Weld quality.

(a) The contractor shall be responsible for determining the inspections needed to insure that, when the welded parts are assembled together to make the bracket, the bracket shall conform to the inspection requirements specified in Section 4.

(b) All weldments shall be free of slag, flux, weld spatter, and other impurities detrimental to either the appearance or strength of the weldment.

(c) Undercut in weldments shall be held to a minimum and shall not be more than 0.031 inch deep.

3.5.4 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than 2 full threads.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:34Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.3.2 Examination.

4.3.2.1 Samples. Samples selected in accordance with 4.3.1 shall be examined for the major and minor defects specified in 4.4.1. AQL shall be 0.4 percent defective for major defects and 4.0 percent defective for minor defects.

4.4 Inspection procedure.

4.4.1 Examination. The bracket shall be examined as specified herein for the following defects:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:34Z
Check the source to verify that this is the current version before use.

Major

101. Bracket not as shown in Figures 1 and 2.
102. Material not as specified.
103. Workmanship not as specified.

Minor

201. Treatment and painting not as specified.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 4.0 percent defective.

104. Containers not as specified.
105. Brackets not cushioned as specified for Level A or Level B.
106. Marking missing, illegible, incorrect, or incomplete for Level A, Level B, or Commercial.

5. PACKAGING

5.1 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

5.1.1 Level A. The brackets shall be folded for minimum cubage and, with the handrail adapter brackets, shall be packed in boxes conforming to PPP-B-601, Overseas Type, or PPP-B-621, Class 2. Cushioning shall be placed between the brackets to prevent movement and resulting damage to the painted surfaces.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:34Z
Check the source to verify that this is the current version before use.

5.1.2 Level B. The brackets shall be packed as specified in 5.1.1 for Level A, except the boxes shall be Domestic Type or class as applicable.

5.1.3 Commercial. The brackets shall be packed in accordance with MIL-STD-1188.

5.2 Marking.

5.2.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.2.2 Commercial. Marking for Commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The bracket is intended for use in erecting bolted steel tanks. The handrail bracket, which is a component of the scaffold bracket, is intended to serve as a mounting for handrail upright supports. NSN 5440-00-319-0118 and NSN 5440-00-510-0327 which are used in Equipment Set 5180-90-CL-N33 apply thereto.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Color required (see 3.3).
(c) Welding process required, when other than at the option of the contractor (see 3.5.2).
(d) Degree of packing required (see 5.1).

6.3 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

Custodian:
Army - ME

Review activity:
DLA - CS

Preparing activity:
Army - ME

Project 5440-A007

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:34Z
Check the source to verify that this is the current version before use.

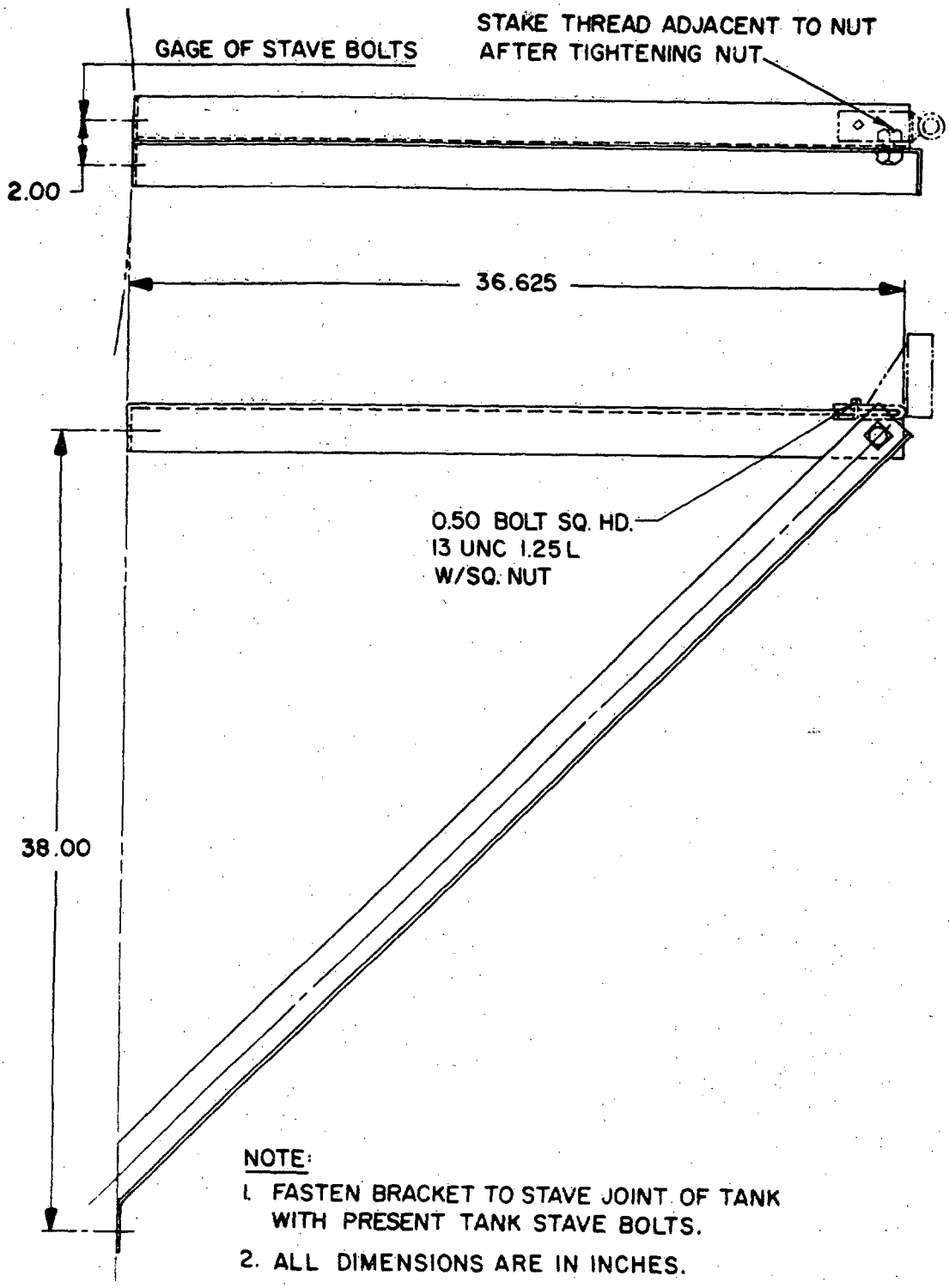
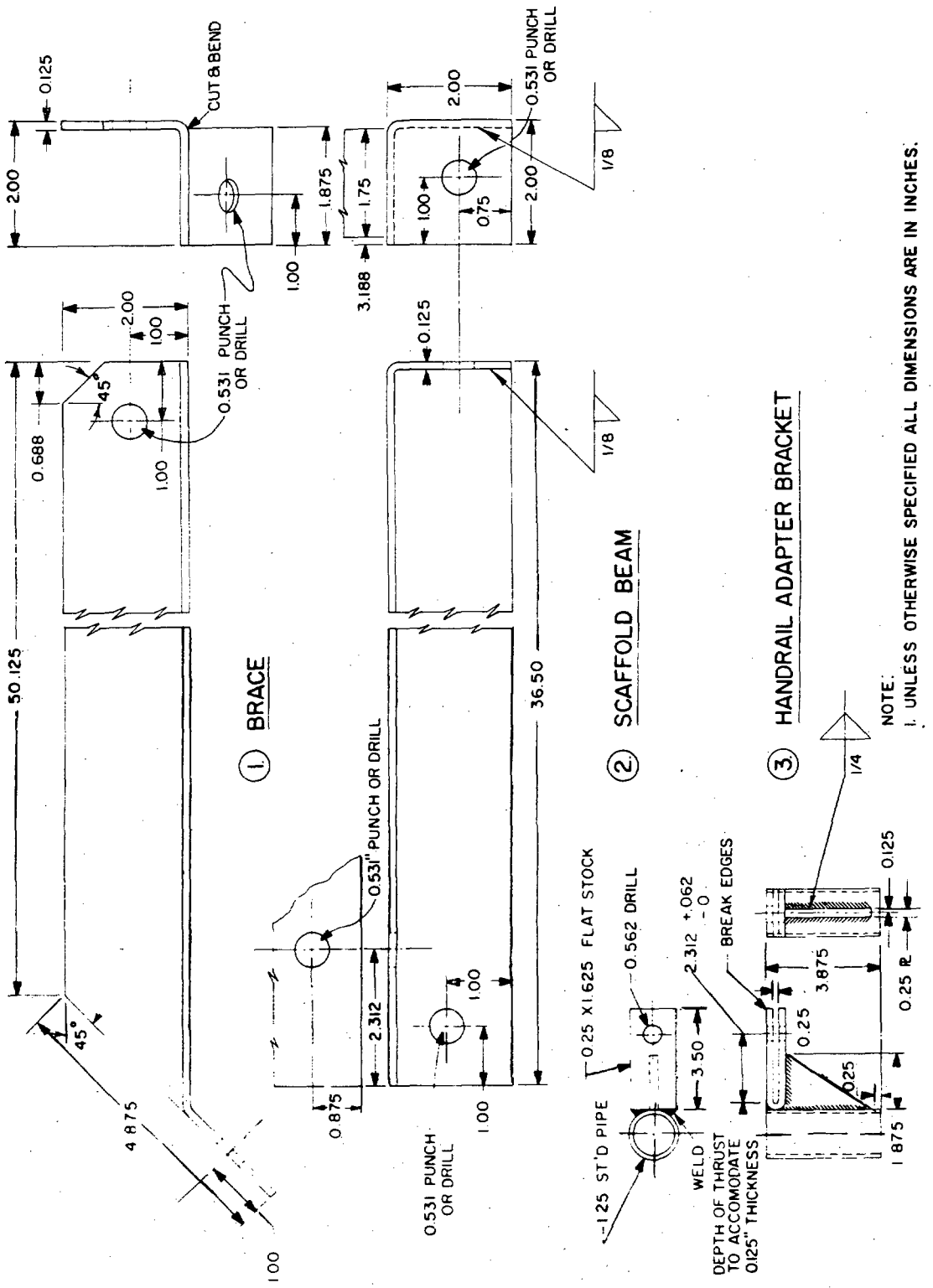


FIGURE 1. Folding scaffold bracket assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:34Z
Check the source to verify that this is the current version before use.

X-2900A



NOTE:
1. UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES.

FIGURE 2. Folding scaffold bracket, details.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:34Z
Check the source to verify that this is the current version before use.